

Science Communication

What is science communication?

Rui Mata, HS 2026

Version: September 17, 2026

Course instructors



Rui Mata



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<http://cds.unibas.ch>

<https://www.superdot.studio>

Learning goals (for the course as a whole)

This course aims to give a broad introduction to science communication. By completing the course, you can expect to...

- Get an overview of theories and principles of science communication
- Learn to critically evaluate and reflect on science communication
- Learn to develop visual communication strategies for different audiences and purposes

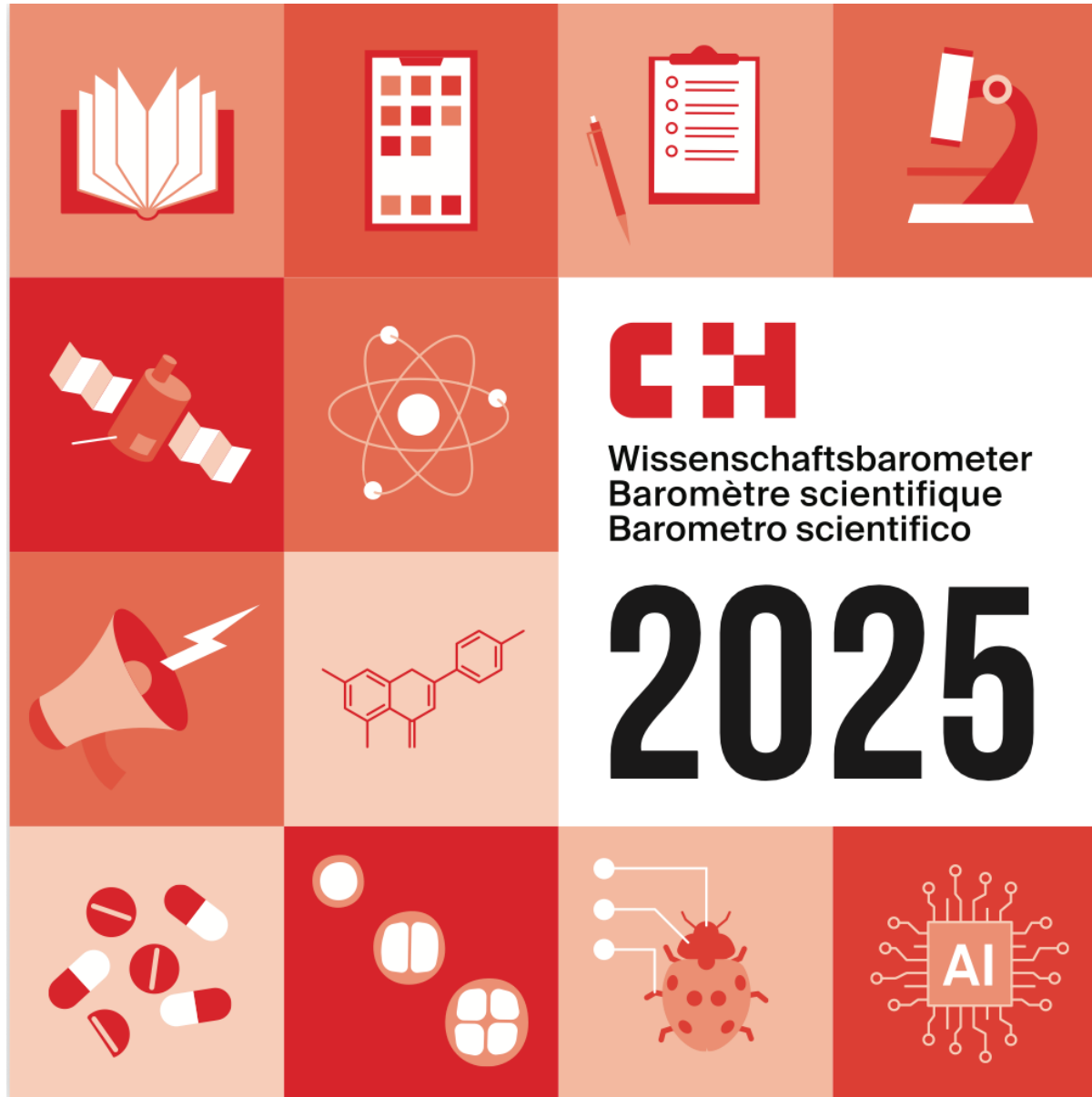
Goals for today

- Recognize the need for effective communication between science and the public, and reflect on the responsibility of scientists (including psychologists) in communicating science effectively
- Discuss the definition of science communication, including various forms and goals
- Become familiarized with the course structure, readings, and website

HOW SHOULD CITIZENS RELATE TO SCIENCE?

Discuss the role of science in society





WIE SOLLTE DAS VERHÄLTNISS VON WISSENSCHAFT, POLITIK UND ÖFFENTLICHKEIT AUSSEHEN?

Die Mehrheit der Schweizer Bevölkerung findet, dass Forschung staatlich unterstützt werden sollte und politische Entscheidungen auf wissenschaftlichen Erkenntnissen beruhen sollten. Zudem sollten Forschende die Öffentlichkeit über ihre Arbeit informieren.

SP- und Grünen-Anhänger befürworten die staatliche Förderung besonders stark. SVP-, SP- und Grünen-Anhänger wollen mitbestimmen, woran geforscht wird. Anhänger der Grünen meinen eher, Forschende wüssten am besten, was gut für die Schweiz ist, während vor allem SVP-Anhänger vermuten, dass Forschende mit Politik und Wirtschaft unter einer Decke stecken.

Basis: 625–1499 Befragte; angegeben sind Mittelwerte, Standardabweichungen zwischen 1.2 und 1.6; Antwortskala: 1 = stimme überhaupt nicht zu bis 5 = stimme voll und ganz zu.

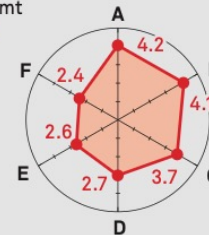
QUELLE RELATION LA SCIENCE DEVRAIT-ELLE ENTRETENIR AVEC LA POLITIQUE ET LE PUBLIC ?

La majorité de la population suisse est favorable au financement public de la recherche et aux décisions politiques fondées sur la science. De plus, les chercheurs devraient informer le public de leurs travaux.

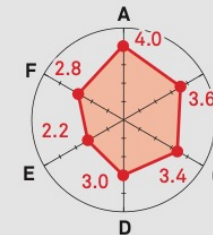
Les partisans du PS et des Verts sont particulièrement favorables au financement public. Les partisans de l'UDC, du PS et des Verts souhaitent avoir leur mot à dire sur les thèmes de recherche. Les partisans des Verts estiment plutôt que les chercheurs savent mieux que quiconque ce qui est bon pour la Suisse, tandis que les partisans de l'UDC soupçonnent surtout les chercheurs d'être de mèche avec les milieux politiques et économiques.

Base : 625–1499 personnes interrogées ; valeurs moyennes indiquées ; écarts-types entre 1.2 et 1.6 ; échelle de réponse : 1 = n'approuve pas du tout à 5 = approuve totalement.

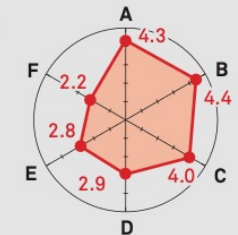
Gesamt
Total



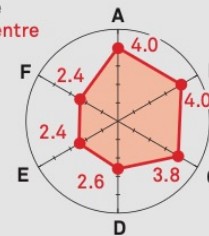
SVP
UDC



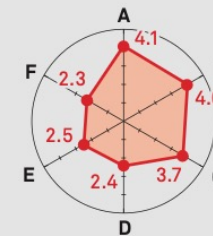
SP
PS



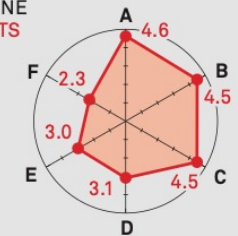
Mitte
Le Centre



FDP
PLR



GRÜNE
VERTS



A Wissenschaftler sollten die Öffentlichkeit über ihre Arbeit informieren.
Les scientifiques devraient informer le public de leurs activités.

B Wissenschaftliche Forschung sollte staatlich unterstützt werden.
La recherche scientifique devrait être soutenue par l'Etat.

C Politische Entscheidungen sollten auf wissenschaftlichen Erkenntnissen beruhen.
Les décisions politiques devraient s'appuyer sur des bases scientifiques.

D Leute wie ich sollten mitentscheiden, zu welchen Themen Wissenschaftler forschen.
Les personnes comme moi devraient prendre part à la décision sur quels thèmes les scientifiques doivent faire des recherches.

E Wissenschaftler wissen am besten, was gut für die Zukunft der Schweiz ist.
Les scientifiques sont les mieux placés pour savoir ce qui est bon pour l'avenir de la Suisse.

F Wissenschaftler stecken mit Politik und Wirtschaft unter einer Decke.
Les scientifiques sont de mèche avec la politique et l'économie.

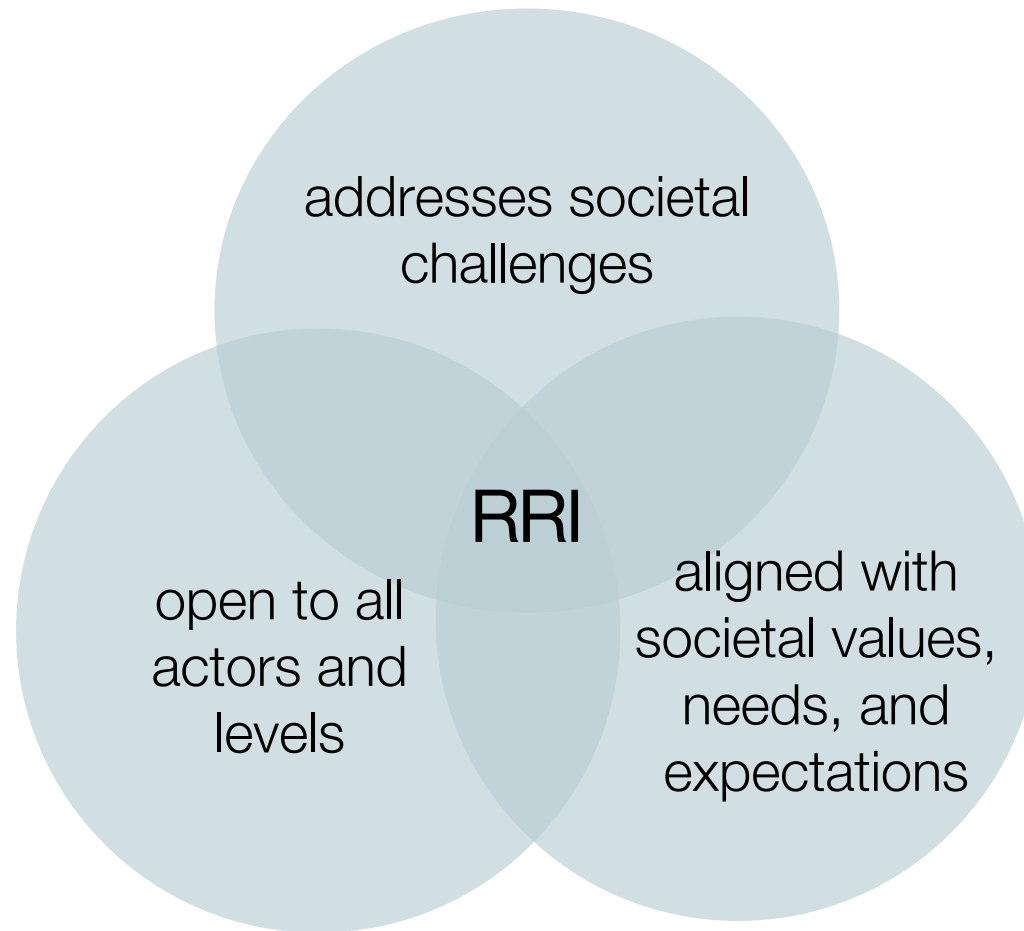
«Baromètre Scientifique Suisse 2022 / WissensCHAftsbarometer Schweiz 2025»

<https://wissenschaftsbarometer.ch/ergebnisse-2025/>

See the comparison to a sample of SCICOM HS26 students [here!](#)

Science with and for society...

Responsible Research & Innovation (RRI)



Bertemes, J. P., Haan, S., & Hans, D. (Eds.). (2024). *50 essentials on science communication*. De Gruyter.
<https://doi.org/10.1515/9783110763577>

WHEN IS SCIENCE COMMUNICATION IMPORTANT?

Identify an example and discuss why science
communication matters





Photo by [Matthew Ball](#) on [Unsplash](#)



Photo by [Mika Baumeister](#) on [Unsplash](#)

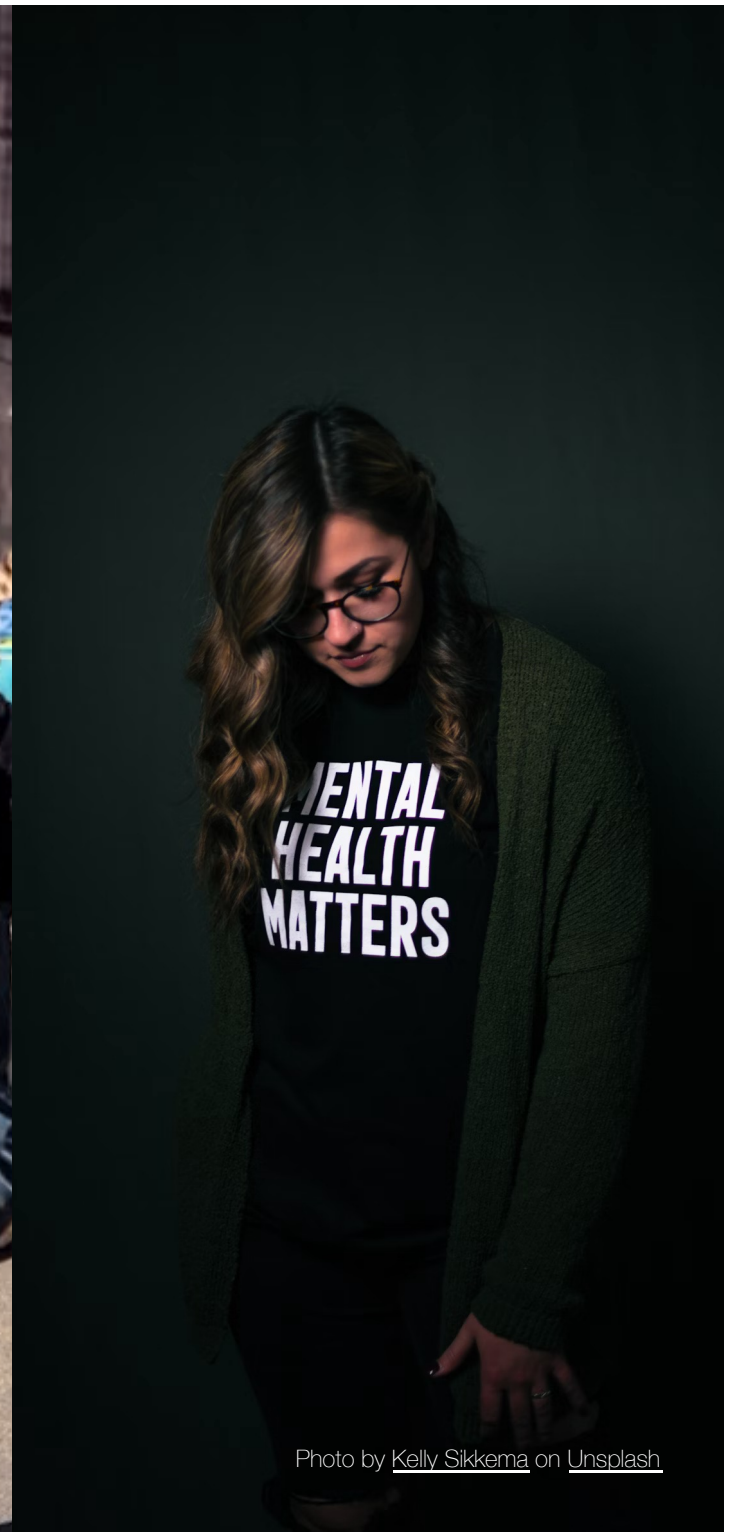


Photo by [Kelly Sikkema](#) on [Unsplash](#)

«three new studies all indicate that combined oral contraceptives (...) are associated with around a two-fold increase in the risk of thromboembolism»

U.K. Committee on Safety of Medicines, 1995

Gigerenzer G, Gaissmaier W, Kurz-Milcke E, Schwartz LM, & Woloshin S. (2007). Helping doctors and patients make sense of health statistics. *Psychological Science in the Public Interest*, 8(2):53-96.

<https://doi.org/10.1111/j.1539-6053.2008.00033.x>

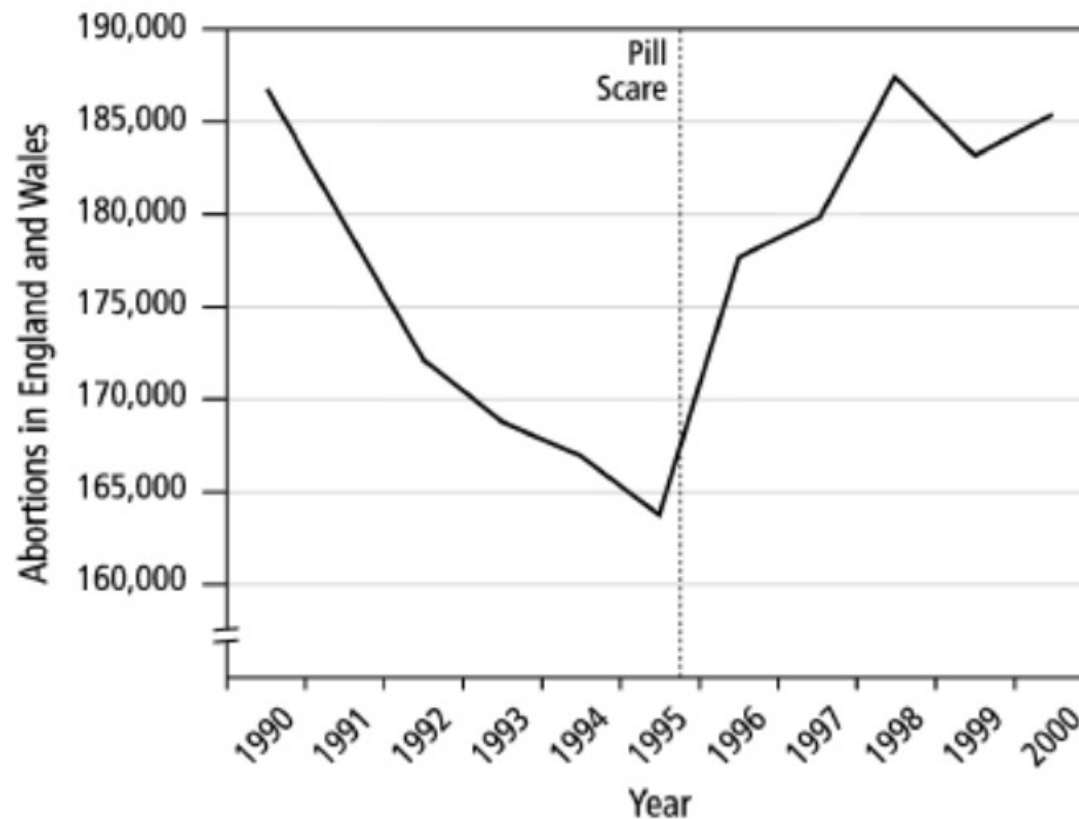


Fig. 1. Reversal of downward trend in number of abortions in England and Wales following the 1995 pill scare.

Gigerenzer G, Gaissmaier W, Kurz-Milcke E, Schwartz LM, & Woloshin S. (20027). Helping doctors and patients make sense of health statistics. *Psychological Science in the Public Interest*, 8(2):53-96.
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2nd gen. contraceptive: 1 out of 7000

3rd gen. contraceptive: 2 out of 7000

the *relative risk* increase is 100%, but the
absolute risk increase was 1 in 7,000...

Science Under Siege

How to Fight the Five Most Powerful Forces that Threaten Our World

Two of the world's most respected scientists reveal the forces behind the dangerous anti-science movement—and show us how to fight back

SCIENCE UNDER SIEGE



**HOW TO FIGHT THE FIVE
MOST POWERFUL FORCES
THAT THREATEN OUR WORLD**

MICHAEL E. MANN

PETER J. HOTEZ

What is science communication?

“We understand science communication as **all forms of communication focused on scientific knowledge or scientific work**, both inside and outside institutionalised science, including its production, content, use, and effects”

FORMS AND GOALS OF SCIENCE COMMUNICATION?

Create a list of examples and
identify categories of forms and goals



Some forms of science communication

Form of Science Communication	Description	Examples
Institutional Science Communication	Communication from scientific institutions or organizations to a non-scientific public.	Universities communicating research to the general public.
Science PR (Public Relations)	Interest-driven communication aimed at building reputation. Part of institutional science communication.	Universities promoting their research to enhance reputation.
Non-institutional Science Communication	Science communication by smaller associations, individuals, or teachers outside institutional settings.	Teachers or individuals sharing their passion for science.
Science Journalism	Reporting on scientific topics by journalists, distinct from institutional science communication.	News articles or reports about scientific discoveries.
Internal Science Communication (Scholarly Communication)	Exchange of scientific information within the scientific community, often during conferences or publications.	Presentations or discussions at academic conferences.
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Goals of science communication

Goals of Science Communication	Description	Examples
Disseminating Information	To inform the public about research activities and findings, fulfilling an ethical obligation.	Researchers sharing findings through public talks or papers.
Building Trust in Science	To enhance the credibility and trustworthiness of researchers and institutions by being visible and transparent.	Open access publications, public outreach by universities.
Preventing Misinformation	To ensure that public discourse is based on academic reality rather than individual beliefs or misinformation.	Clear communication on topics like climate change or vaccines.
Encouraging Critical Thinking	To help the public understand that uncertainty and controversy are natural parts of the scientific process.	Engaging in dialogue about scientific ambiguity and debates.
Supporting Informed Public Decision-Making	To provide the public with accurate information, allowing for informed, democratic decisions on scientific issues.	Public debates or campaigns around science-related policies.
Engaging in Public Dialogue	To create a two-way communication where scientists also consider societal needs and interests in their research.	Public forums or discussions where scientists and public interact.
Creating a Competitive Edge	To differentiate institutions or researchers by showcasing scientific achievements, though potentially controversial.	Researchers or universities promoting breakthroughs to attract funding.

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What is science communication?

SCIENCE COMMUNICATION (SciCom) may be defined as the use of appropriate skills, media, activities, and dialogue to produce one or more of the following personal responses to science (the vowel analogy)

Awareness, including familiarity with new aspects of science

Enjoyment or other affective responses, e.g. appreciating science as entertainment or art

Interest, as evidenced by voluntary involvement with science or its communication

Opinions, the forming, reforming, or confirming of science-related attitudes

Understanding of science, its content, processes, and social factors

Science communication may involve science practitioners, mediators, and other members of the general public, either peer-to-peer or between groups.

Figure 1. The AEIOU definition of science communication. This definition clarifies the purpose and characteristics of science communication and provides a basis for evaluating its effectiveness

Burns, T. W., O'Connor, D. J., & Stocklmayer, S. M. (2003). Science communication: A contemporary definition. *Public Understanding of Science*, 12(2), 183–202. <https://doi.org/10.1177/09636625030122004>

Summary

- **Responsible Research & Innovation:** aspirational, collaborative approach to R&I that addresses societal challenges, involves all actors at various levels, and aligns with societal values, needs, and expectations.
- **Science communication:** all forms of communication focused on scientific knowledge or scientific work; the use of various skills, media, and activities to engage diverse audiences in science, fostering **A**wareness, **E**njoyment, **I**nterest, **O**pinions, and **U**nderstanding of science through dialogue and interaction of multiple actors (AEIOU mnemonic).
- **Science communication entails multiple forms and goals:** ranging from institutional (e.g., universities) and non-institutional efforts (e.g., small associations, individuals), science journalism, and others, emphasizing the diversity of approaches used to engage different audiences in science-related content; each form can serve different goals, such as dissemination and knowledge transfer, but also critical thinking (because scientific results are not always clear cut), or reputation building (which can pose challenges to unbiased, truthful communication).

Course website and structure

Session information

Sessions take place Thursdays, 8.15–9.45, Biozentrum, Hörsaal U1.131.

#	Date	Topic	Instructor(s)	Material	Exercise
1	17.09.2026	What is science communication?	Mata	pdf	
2	08.10.2026	Models and elements of science communication	Mata	pdf	
3	15.10.2026	Scientific uncertainty and trust in science	Mata	pdf	
4	22.10.2026	Guidelines for science communication	Mata	pdf	
5	29.10.2026	Science communication in practice	Mata	pdf	
6	05.11.2026	Why do we visualize?	Hil/Lachenmeier		
7	12.11.2026	Structured content	Hil/Lachenmeier		A
8	19.11.2026	Content dimensions and categorization	Hil/Lachenmeier		B
9	26.11.2026	The identity of a dot	Hil/Lachenmeier		C
10	03.12.2026	Networks and relationships	Hil/Lachenmeier		D
11	10.12.2026	Visual language and cultural context & Areas and hierarchies	Hil/Lachenmeier		E
12	17.12.2026	Exam			

https://matarui.github.io/SCICOM_HS26/

Recap of last session

- Recognize the need for effective communication between science and the public, and reflect on the responsibility of scientists (including psychologists) in communicating science effectively
- Discuss the definition of science communication, including various forms and goals
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WHO'S GOT THE ANSWER?

According to the principle of responsible research and innovation...

[Select all the correct answers]

• A: Science is open to all

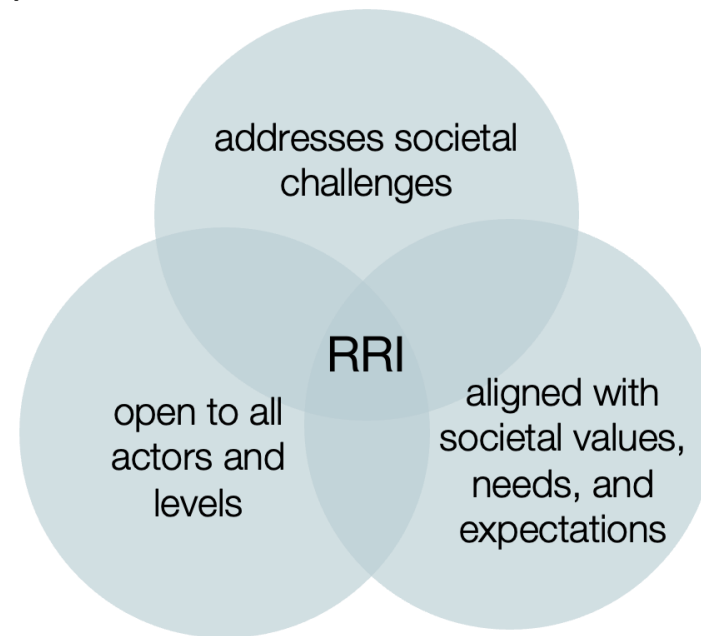
• B: Science is mostly concerned with human health

• C: Science is efficient and avoids risks

• D: Science is aligned with societal needs



Responsible Research & Innovation (RRI)



According to the principle of responsible research and innovation...

[Select all the correct answers]

• **A: Science is open to all**

• B: Science is mostly concerned with human health

• C: Science is efficient and avoids risks

• **D: Science is aligned with societal needs**

In the AEIOU definition of science communication...
[Select all the correct answers]

• A: A stands for AWE

• B: E stands for EXPLICIT

• C: I stands for INTEREST

• D: O stands for OPINIONS



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In the AEIOU definition of science communication...
[Select all the correct answers]

• A: A stands for AWE

• B: E stands for EXPLICIT

• C: I stands for INTEREST

• D: O stands for OPINIONS

An example of internal science communication is when...
[Select all the correct answers]

• A: a scientist presents findings at an academic conference

• B: a scientist discusses preliminary results with a colleague

• C: a scientist gives a public talk at a community event

• D: a scientist helps write a press release about their work



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☐ C: a scientist gives a public talk at a community event

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An example of institutional science communication is when...
[Select all the correct answers]

- A: UNIBAS releases a new episode of UNISONAR
- C: UNIBAS issues a press release about a new paper

- B: a scientist shares opinions about science in a personal blog
- D: UNIBAS sends a science newsletter to its alumni



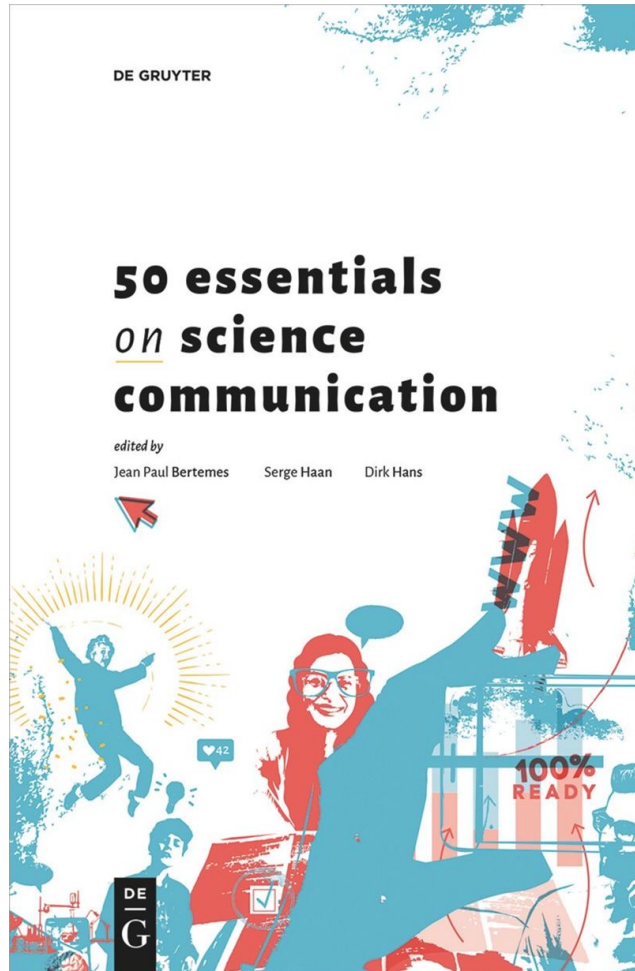
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Knowledge Transfer	Communication between science and industry or societal actors, often for practical applications.	Collaborations between universities and industry.

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[Select all the correct answers]

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- ☒ **C:** UNIBAS issues a press release about a new paper

- ☐ **B:** a scientist shares opinions about science in a personal blog
- ☒ **D:** UNIBAS sends a science newsletter to its alumni

Course readings



Grading

The final grade for the course will be based on two components:

- **Exam (55%):** At the end of the semester, a multiple-choice exam will test your understanding of core concepts covered in lectures and readings.
- **Assignments (45%):** A portion of the grade will come from practical assignments, which involve applying the principles learned to specific applications. Both timely submission and the quality of work will be key factors in determining PASS/FAIL for each assignment. Assignments should be submitted via ADAM by the designated deadlines. More information will be available on the course website.